recidivism rate. Remember that tasks that are completed can have a maximum recidivism of 50% or .5, because they would have moved backward as many times as they moved forward. To take the output of the recidivism formula and equate that to a number between 0 and 25 where 0 is the worst and 25 is the best, they normalized the result by multiplying recidivism by 50 (25 * 2). Using that formula the highest possible rate of recidivism of completed tasks would be 0, and the lowest possible result would be 25.

As a reminder, we used the following formula earlier for recidivism:

```
Recidivism = Backwards Tasks / (Forward Tasks + Backwards Tasks)
```

Or

```
Recidivism = 25 – ((bN / (fN + bN)) *50)
```

With that formula some example inputs and outputs are shown in table 7.4..

**Table 7.4. Example inputs and outputs for a normalized recidivism**

| Input | Output | Result |
|---|---|---|
| bN = 5, fN = 100 | 22.62 | Not bad |
| bN = 100, fN = 100 | 0 | The worst possible output |
| bN = 0, fN = 125 | 25 | Perfect score |

## Normalizing escaped defects

An escaped defect was a bug that wasn't caught in the release process; it was found after the release was considered a success. Finding any escaped defects should make the team stop and find out what happened so they could continue to improve their automated test and release. Because of this even a single escaped defect should count against the total. The formula for escaped defects was pretty simple: multiply them and subtract from 25. As